

Research - Client/Server Architecture

By Nicolas Barrios, Andrew Mendez, Jayden Baltazar, Jennyfer Hernandez Rosas

Use Cases

1. Login: - Completely Working

Description: System authenticates the user's inputted username and password before giving access to the system

Pre-condition: The Client and Server are both running concurrently.

Post-condition: User is either granted access to the server or isn't

Valid Username/Password

1. User enters a valid Username - NicoBarr and valid password - strongMAN
2. Client sends login credentials to system
3. System checks the database for credentials - Success
4. System sends successful login response
5. System grants access to user
6. END

Next Use Case: Connect

Invalid Password

1. User enters a valid username - NicoBarr and invalid password - strongMA
2. Client sends login credentials to system
3. System checks the database for credentials - Fail
4. System sends fail login response
5. System increments error count by 1
6. System doesn't grant access to user
7. Client displays Invalid password
8. END

Next Use Case: Login

Invalid Username

1. User enters a invalid Username - NicoBarr and valid password - strongMAN
2. Client sends login credentials to system
3. System checks the database for credentials - Fail
4. System sends fail login response
5. System doesn't grant access to user
6. Client displays username doesn't exist
7. END

Next Use Case: New user registration

Account Lockout

1. User enters a valid email - NicoBarr and invalid password - strongMA for a 3rd time
2. Client sends login credentials to system
3. System checks the database for credentials - Fail
4. System sends fail login response

5. System doesn't grant access to user
6. System increments error count is 3
7. System doesn't grant access to user
8. Client displays Invalid password and tells user is locked out
9. END

Next Use Case: Password Recovery

2. Connect: - Completely Working

Description: System authenticates the user's inputted IP address before connecting to the server

Pre-condition: The Client and Server are both running concurrently.

Post-condition: User is either connected to the server or isn't

Valid IP Address

1. User supplies and enters a server IP address - 127.0.0.1
2. Server sends the IP address to the Client
3. Client verifies the IP address is valid - Success
4. System sends a success response
5. System connects to server with the provided IP address
6. System connection successful
7. END

Next Use Case: Disconnect

Invalid IP Address

1. User supplies and enters a server IP address - 127.0.0.&
2. Server sends the IP address to the Client
3. Client verifies that the IP address is valid - Fail
4. System sends a failed response
5. System does not connect to the server to the provided IP address
6. System connection unsuccessful
7. End

Next Use Case: Disconnect

3. Disconnect: - Completely Working

Description: System disconnects from the user's inputted IP address

Pre-condition: The Client and Server are both running concurrently.

Post-condition: User is either disconnected or not disconnected to the server

Successful Disconnection

1. User selects disconnect from server
2. Client is disconnected from the server
3. Client is still running
4. END

Next Use Case: Connect

Failure to Disconnect

1. User selects disconnect from server
2. Client fails to disconnected from the server (e.g. server timeout)
3. Client is still connected to the server

4. Systems sends response → “Client is still connected to server”
5. END

Next Use Case: Disconnect

Random Disconnection

1. Client loses wifi connection
2. Client is disconnected from server
3. END

Next Use Case: Connect

4. Password Recovery: - Completely Working

Description: System allows the user to reset their password through their Username.

Pre-condition: User inputs wrong password 3 times and is locked out of the server then presses password reset button

Post-condition: User is either receives the email with one-time passcode or doesn't

Existing Email

1. User types in recipient Username - JayBalz
2. Client sends Username to System
3. System checks if there is an email tied to the username - Success
4. System sends email a one-time passcode
5. User receives the email with the one-time passcode
6. END

Next Use Case: Login

Non-Existent Email

1. User types in Username- JayBallz
2. Client sends Username to system
3. System checks if there is an email tied to the username - Fail
4. System doesn't email a one-time passcode
5. Client displays “Username Not Found” message
6. END

Next Use Case: Password Recovery

Server Error

1. User types in Username -JayBalz
2. Client packages email content and sends to system
3. System attempts to process Username -Fail (e.g., server timeout)
4. System sends error response
5. Client displays “Failed to Send one time code. Please Try Again Later.”
6. END

Next Use Case: Logout

5. Logout: - Completely Working

Description: System ends User's session and safely closes any open connections

Pre-condition: The client is connected to the server and the user is authenticated

Post-Condition: User session ends and the system returns to login screen

User Auto Logout

1. User leaves Client running from sudden shutdown or inactivity.
2. User/s gets logged out of the system
3. Client automatically disconnects from inactivity
4. Client stops running.
5. End.

Next Use Case: Login

Manual User Logout

1. User selects logout.
2. User gets logged out of the system.
3. Client disconnects from the server.
4. End.

Next Use Case: Login

6. New User Registration: - Completely Working

Description: User creates a account to login and credentials are kept within the database

Pre-conditions:

- The Client and Server are both running concurrently
- The User does not have an account

Post-condition: User creates a new username, password, and email address associated with the username

New Account Process (Valid Credentials)

1. User requests the system to make a new account
2. Systems requests a username and the User inputs:
 - “AnthonyM”
3. System requests a password and the User inputs with the required format to make the password (8 character minimum:
 - a. At least 1 uppercase and lowercase letter
 - b. At least 1 digit
 - c. At least 1 special character)
 - “AndrewM1234#”
4. System checks if the password fulfilled the required format and determines the password is valid
5. System requests a email address and the User inputs:
 - “Andrew@gmail.com”
6. System determines the login credentials are valid and stores them into the database
7. END

Next Use Case: Login

New Account Process (Invalid Credentials)

1. User requests the system to make a new account
2. Systems requests a username and the User inputs:
 - “AnthonyM”
3. System requests a password and the User inputs with the required format to make the password (8 character minimum:

- a. At least 1 uppercase and lowercase letter
 - b. At least 1 digit
 - c. At least 1 special character)
- "am"
4. System checks if the password fulfilled the required format and determines the password is invalid
5. System displays a message saying "The password does not fulfill the password requirements, please check the requirements and try again"
6. User inputs:
7. "AndrewM123#"
8. System checks if the password fulfilled the required format and determines the password is valid
9. System requests a email address and the User inputs:
 - "Andrew@gmail.com"
10. System determines the login credentials are valid and stores them into the database
11. END

Next Use Case: Login

7. Shutdown: - Completely Working

Description: User shutdown their client

Shutdown While Logged Out

Pre-conditions: User has the system running and is logged out

1. User requests to shutdown the system
2. System terminates the client
3. END

Post Condition: System is terminated for the user

Next Use Case: Login

Shutdown While Logged In

Pre-conditions: User has the system running and is logged in

1. User requests to shutdown the system
2. System terminates the client
3. System logs out the account that was logged in
4. END

Post Condition: System is terminated for the user

Next Use Case: Login

Unexpected Shutdown While Logged In

Pre-conditions: User has the system running and is logged in

1. User experiences an unexpected shutdown (e.g. computer runs out of charge)
2. System terminates the client
3. System logs out the account that was logged in
4. END

Post Condition: System is terminated for the user

Next Use Case: Login

Shutdown on the Server GUI

Pre-conditions: User has the system running

1. User on the server side closes the GUI.
2. System terminates the server GUI
3. END

Post Condition: System is terminated for the user

Next Use Case: Login

8. Display Client/Server Info: - Completely Working

Description: Admin on the server side clicks button to view all information

Pre-condition: The Server is running concurrently.

Post-condition: Admin on the server sees information about the Client Side

Display all Information

1. Admin clicks on the query button
2. System checks the database for the number of registered users, number of logged in users, and number of locked out users.
3. System saves that information
4. System checks which users are logged in and the number of current users on the client.
5. System saves that information
6. System packs all the information
7. System displays the information to the Admin
8. END

Next Use Case: **Shutdown**